

Doohyun Park, Ph.D.

Research Scientist

Seoul, Republic of Korea
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RESEARCH AREAS

1. Representation Learning and Label-Efficient Learning in Medical Imaging
2. Generalization and Validation of Medical AI in Real-World Clinical Settings
3. Multimodal Medical Data Modeling for Diagnosis and Prognosis

EDUCATION

Yonsei University

Ph.D. School of Electrical and Electronic Engineering

Dissertation: *Artificial Intelligence-based Preoperative Prediction of Axillary Lymph Node Metastasis in Breast Cancer using Whole Slide Images*

Mar. 2016 – Feb. 2024
Seoul, Republic of Korea

Yonsei University

B.S. School of Electrical and Electronic Engineering

Mar. 2012 – Feb. 2016
Seoul, Republic of Korea

WORK EXPERIENCE

VUNO Inc.

Research Scientist (Military Obligation)

- Developed a self-supervised foundation model for medical imaging to enhance downstream task performance and data efficiency.
- Designed clinical validation protocols and performance evaluation frameworks aligned with real-world deployment requirements.
- Implemented noisy-label detection and dataset refinement strategies to improve training robustness.
- Optimized model architecture for reduced computational cost and practical deployment.
- Conducted proof-of-concept integration of multimodal medical LLMs into product-oriented pipelines.

Dec. 2023 – now
Seoul, Republic of Korea

VUNO Inc.

Visiting Researcher

- Led two independent research projects, each resulting in a first-author peer-reviewed publication.
- Developed a weakly supervised deep learning model for medical image abnormality detection and classification.
- Developed an AI model for COVID-19 severity assessment using multi-center clinical data.

Aug. 2021 – July 2022
Seoul, Republic of Korea

SELECTED PUBLICATIONS

1. Frequency-Balanced Retinal Representation Learning with Mutual Information Regularization
S Lee, S Kang, I Park, G K won, J Baek, D Park[†]
[†] Corresponding author
[2026 ICLR, International Conference on Learning Representations](#)
2. Task-Agnostic Noisy Label Detection via Standardized Loss Aggregation
I Park, D Park[†]
[†] Corresponding author
[2026 ISBI, IEEE International Symposium on Biomedical Imaging](#)
3. A Comparative Study of Machine Learning and Deep Learning for Out-of-Distribution Detection
J Baek, S Lee, G Kwon, D Park[†]
[†] Corresponding author
[2026 ISBI, IEEE International Symposium on Biomedical Imaging](#)
4. Multimodal AI model for preoperative prediction of axillary lymph node metastasis in breast cancer using whole slide images
D Park, YM Lee, T Eo, HJ An, H Kang, E Park, YJ Cha, H Park, D Kwon, SY Kwon, HR Jung, SJ Shin, H Park, Y Lee, S Park, JM Kim, SE Choi, NM Cho[†], D Hwang[†]
[2025 npj Precision Oncology \(JCR top 11.7%\)](#)
5. A Comprehensive Review of Performance Metrics for Computer-Aided Detection Systems
D Park
[2024 Bioengineering \(JCR top 40.7%\)](#)
6. Deep Learning-based Slice Thickness Reduction for Computer-Aided Detection of Lung Nodules in Thick Slice CT
J Jeong*, D Park*, JH Kang, M Kim, HY Kim, W Choi, J Kim, SY Ham
* Equally Contributed
[2024 Diagnostics \(JCR top 17.0%\)](#)
7. Weakly-Supervised Deep Learning for Multi-Label Classification of Vertebral Compression Fracture in CT
E Choi*, D Park*, G Son, S Bak, T Eo, D Youn, D Hwang
* Equally Contributed
[2024 European Radiology \(JCR top 12.4%\)](#)
8. Development and Validation of A Hybrid Deep Learning-Machine Learning Approach for Severity Assessment of COVID-19 and Other Pneumonias
D Park, R Jang, MJ Chung, HJ An, S Bak, E Choi, D Hwang
[2023 Scientific Reports \(JCR top 18.3%\)](#)
9. Importance of CT Image Normalization in Radiomics Analysis: Prediction of 3-Year Recurrence-Free Survival in Non-Small Cell Lung Cancer
D Park, D Oh, M Lee, SY Lee, KM Shin, J-SG Jun, D Hwang
[2022 European Radiology \(JCR top 11.5%\)](#)

OTHER PUBLICATIONS

1. Prompt-Induced Bias from One-Shot Multimodal Large Language Models Prompting in Medical Imaging
I Park, J Baek, D Park
[2026 ISBI, IEEE International Symposium on Biomedical Imaging - abstract](#)
2. Prompt Dominance in Medical Multimodal Large Language Models
I Park, J Baek, D Park
[2026 ISBI, IEEE International Symposium on Biomedical Imaging - abstract](#)

3. Interpretable Multimodal Transformer for Prediction of Molecular Subtypes and Grades in Adult-type Diffuse Gliomas
Y Byeon, YW Park, S Lee, [D Park](#), H Shin, K Han, JH Chang, SH Kim, SK Lee, SS Ahn, D Hwang
[2025 npj Digital Medicine](#)
4. Multi-center Validation of Pulmonary Nodule Classification Model for Lung Cancer Screening
[D Park](#), JH Kang, C Park
[2025 Preprints](#)
5. Attention-based Interpretable Deep Learning with Radiomic Features for Pulmonary Nodule Classification
[D Park](#), N Lee, S Lim
[2025 MIDL](#), Medical Imaging with Deep Learning – short paper track
6. Foundation Model-based Unsupervised CT Kernel Conversion for Standardizing Emphysema Quantification
[D Park](#), J Kang, J Jeong
[2025 ECR \(Oral Presentation\)](#), European Congress of Radiology
7. Evaluation Metrics for Computer-Aided Detection Systems in Medical AI: What You Need to Know
[D Park](#)
[2025 ECR](#), European Congress of Radiology
8. Deep Learning-based Lesion Segmentation for Patients with Usual Interstitial Pneumonia in High-Resolution CT: A pilot study
[D Park](#), J Kang, J Jeong, KM Moon
[2025 JRC](#), Japan Radiology Congress
9. Fully Automated Segment-Anything-Model for Robust Lobe Segmentation in Patients with Interstitial Lung Disease
S Lim, [D Park](#), J Jeong, C Park, KM Moon
[2025 JRC](#), Japan Radiology Congress
10. Deep Learning-based Slice Thickness Reduction for Lung Nodule Detection of Thick Slice Chest CT
J Jeong, [D Park](#), JH Kang, M Kim, HY Kim, W Choi, SY Ham
[2025 JRC](#), Japan Radiology Congress
11. Nodule Type Classification for Lung Cancer Screening with CT
[D Park](#), J Kang, C Park, J Kim
[2024 WCLC](#), World Conference on Lung Cancer
12. A Standardized Performance Evaluation Metric for Chest CT Nodule Detection
[D Park](#), C Park, J Kang, J Kim
[2024 ESTI](#), European Society of Thoracic Imaging
13. A Multi-center Retrospective Study of Nodule Type Classification using Artificial Intelligence for Lung-RADS Scoring with CT
[D Park](#), J Kang, C Park, J Jeong
[2024 RSNA](#), Radiological Society of North America
14. Quantity or Certainty: Can Ambiguously Annotated Data Improve Lung Nodule Detection Performance?
C Park, J Kang, [D Park](#), J Jeong
[2024 RSNA](#), Radiological Society of North America
15. LLM-guided Multi-modal Multiple Instance Learning for 5-year Overall Survival Prediction of Lung Cancer
K Kim, Y Lee, [D Park](#), T Eo, D Youn, H Lee, D Hwang
[2024 MICCAI](#), International Conference on Medical Image Computing and Computer Assisted Intervention
16. Unified Diffusion model for Multi-contrast Ensembling Synthesis
Y Lee, Y Shin, [D Park](#), G Son, T Eo, D Hwang
[2024 ISMRM \(Oral Presentation\)](#), International Society for Magnetic Resonance in Medicine

17. Deep Learning Algorithm for Prediction of Molecular Subtypes and Grades in Adult-type Diffuse Gliomas: According to the 2021 WHO Updates
Y Byeon, YW Park, S Lee, H Shin, [D Park](#), SS Ahn, S-K Lee, D Hwang
[2024 ISMRM \(Oral Presentation\)](#), [International Society for Magnetic Resonance in Medicine](#)
18. Deep Learning-based Joint Effusion Classification in Adult Knee Radiographs: A Multi-Center Prospective Study
H Won, HS Lee, D Youn, [D Park](#), T Eo, W Kim, D Hwang
[2024 Diagnostics](#)
19. M3F: Multi-Field-of-View Feature Fusion Network for Aortic Vessel Tree Segmentation in CT Angiography
Y Byeon, H Kim, K Kim, [D Park](#), E Choi, D Hwang
[3rd Place](#), MICCAI Grand Challenge (SEG.A – Segmentation of the Aorta), Technical Paper
[2023 LNCS \(Best Paper Award\)](#), [Lecture Notes in Computer Science](#)
20. Foundation Models in Healthcare: Applications, Challenges, and Future Directions
H Won, [D Park](#), T Eo, Y Lee, D Hwang
[2023 CICS](#), [Conference on Information and Control Systems](#)
21. Iterative Cross-Domain Deep-Learning Approach for Reconstructing Undersampled Radial MRI
[D Park](#), T Eo, T Kim, J Jang, D Hwang
[2018 ISMRM](#), [International Society for Magnetic Resonance in Medicine](#)
22. Deep Sinogram Learning for Radial MRI: Comparison with k-space and Image Learning
T Kim, T Eo, [D Park](#), Y Jun, D Hwang
[2018 ISMRM](#), [International Society for Magnetic Resonance in Medicine](#)
23. Correction of severe beam-hardening artifacts via a high-order linearization function using a prior-image-based parameter selection method
D Oh, S Kim, [D Park](#), S Choi, H Song, Y Choi, S Moon, J Baek, D Hwang
[2018 Medical Physics](#)
24. Beam Hardening Correction Using Length Linearization
D Oh, S Kim, [D Park](#), D Hwang
[2017 SPIE Medical Imaging](#), [International Society for Optics and Photonics](#)

INTERNATIONAL PATENTS

1. Method of predicting prognosis of patient with adenocarcinoma using image feature
[2025 US](#), [Granted No. US12500003B2](#)
2. Makeup evaluation system and operating method thereof
[2023 US](#), [Granted No. US11748980B2](#)
3. Cosmetic evaluation system and operation method thereof
[2022 China](#), [Granted No. CN110235169B](#)
4. Makeup evaluation system and its operation method
[2022 Japan](#), [Granted No. JP7020626B2](#)
5. Makeup evaluation system and operating method thereof
[2021 US](#), [Granted No. US11113511B2](#)
6. Method of predicting prognosis of patient with adenocarcinoma using image feature
[2024 EPO](#), [Application No. EP4218032A4](#)
7. Method of predicting prognosis of patient with adenocarcinoma using image feature
[2023 China](#), [Application No. CN116325019A](#)
8. Makeup evaluation system and operating procedures for it
[2020 EPO](#), [Application No. EP3579176A4](#)

DOMESTIC PATENTS

1. Method for Predicting Prognosis in Cancer Patient Using Radiomic Feature
[2025 Korea](#), [Granted No. 10-2758775](#)
2. Method for outputting segmentation information for an image and device therefor
[2024 Korea](#), [Application No. 10-2024-0176698](#)
3. Method for Predicting Prognosis in Lung Cancer Patient using Clinical Information and Gene Polymorphism Information
[2021 Korea](#), [Granted No. 10-2305806](#)
4. Make-up evaluation system and operating method thereof
[2020 Korea](#), [Granted No. 10-2066892](#)
[Technology transfer with ₩20,000,000](#)
5. Device and Method for Reconstructing Computed Tomography Image
[2019 Korea](#), [Granted No. 10-2039472](#)

TEACHING EXPERIENCE & INVITED TALK

1. Global Engagement & Empowerment Forum on Sustainable Development (GEEF)
Topic: *Ensure Healthy Lives by Predicting the Prognosis of Cancer Patients Using Medical Images*
[2023 GEEF](#), [about United Nations \(UNs\) Sustainable Development Goals \(SDGs\)](#)
2. Yonsei University Graduate School
Topic: *Clinical Applications and Their Technologies Based on Deep Learning*
[2023 Guest lecturer](#)
3. Teaching Assistant - Electrical and Electronic Engineering Capstone Design
Topic: Deep learning-based segmentation in CT
[2018 Yonsei University](#)
4. Teaching Assistant - Digital Signal Processing
[2017 Yonsei University](#)
5. Teaching Assistant - Electrical and Electronic Engineering Capstone Design
Topic: Iterative CT Reconstruction Algorithm
[2016 Yonsei University](#)
6. Teaching Assistant - Electrical and Electronic Engineering Capstone Design
Topic: Feature Selection Algorithm
[2016 Yonsei University](#)